TMS PROJECT

# Flight alerts for a travel agency: warn clients two hours before a delay or cancellation

NYC flights 2023 (JFK, LGA, EWR) · 435,352 flights


Team

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# Warn clients two hours before a flight goes wrong, and turn a bad surprise into a managed one

## 1 in 4

flights were late or cancelled, Jan-Sep 2023 (25.7% of 330,883) [01]

## 2 hours

of notice: time to warn clients and prepare alternatives

## 2 alerts

tested once on Oct-Dec flights the model never saw [07]

**The ask:** approve a pilot of two alerts, with a person checking every one.

Data: nycflights23, all flights from JFK, LGA and EWR in 2023 (435,352 flights).

Sources: notebooks 01, 07

# Three questions, and two traps in the data that we had to fix

## The three questions

**Q1** Which flights will be late or cancelled? **(Alert 1)**

**Q2** Who is most and least reliable, and how much is weather? **(findings)**

**Q3** Which flights will be cancelled? **(Alert 2)**

435,352 flights, 5 files, no row removed. [01]

## The two traps

**1. Weather is in UTC,** flights are in New York time.

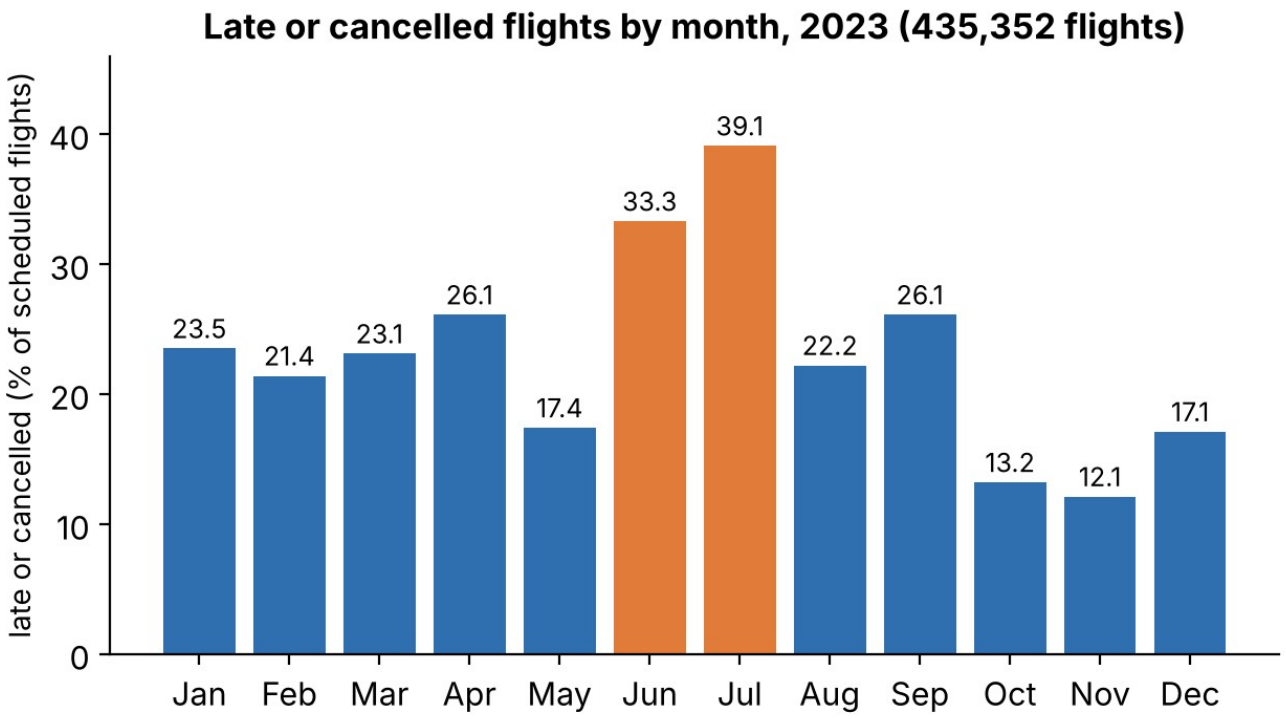
We converted first. A literal join would change the score by only +0.0005. [06]

**2. No tail number = cancelled** (1,913 of 1,913 flights). [01]

We fill plane data with a random real aircraft of the same airline: leak +0.012, not +0.206. [05]

Sources: notebooks 01, 05, 06

# 1 flight in 4 is late or cancelled, and July is the worst month




Orange = above 30%. About 36,000 flights a month. [04]

June and July were not wetter or foggier: a few bad days drive them, and the cause is **not explained by the available data**. [04]

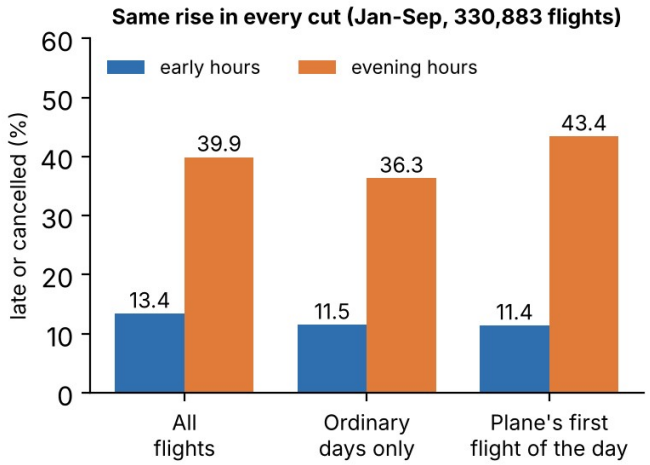
| Metric, whole year [04] | Value | Out of |
|---|---|---|
| M1 On time (delay ≤ 15 min) | 79.03% | departed flights |
| M2 Mean delay | 13.84 min | departed flights |
| M3 Cancelled | 2.47% | scheduled flights |
| M4 More than 1 h late | 8.41% | departed flights |
| M5 Late or cancelled | 22.91% | scheduled flights |

**Today** a client finds out at the airport. We want them to know two hours earlier.

Sources: notebook 04 (whole year, 435,352 flights)

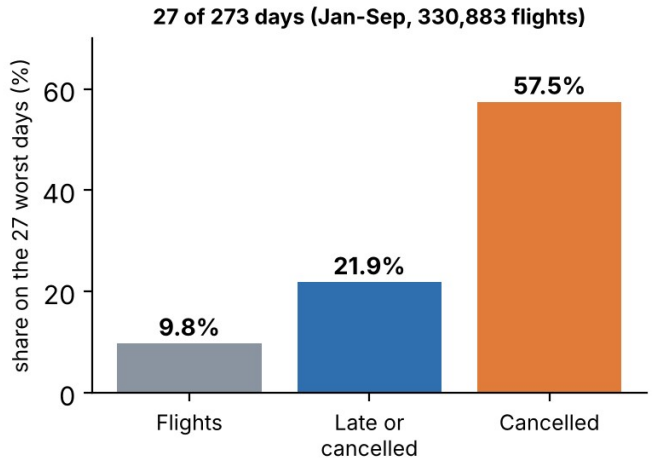
# Three findings: risk builds through the day, a few days do the damage, and airport state beats weather

## 1. Risk builds through the day



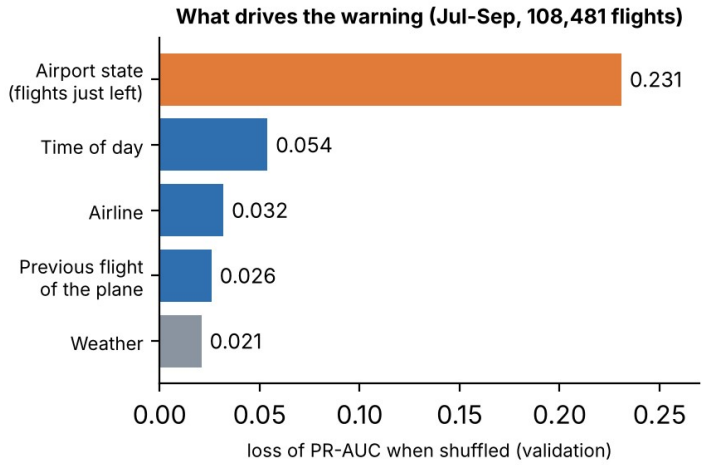

**13.4%** → **39.9%** late or cancelled, early vs evening (gap 26.5 points, interval 24.2 to 28.9). [02]

## 2. A few bad days do the damage




**27 days out of 273** hold **57.5%** of all cancellations. [02]

## 3. Airport state beats weather




Airport state **0.231** against weather **0.021**. Rain adds 14.5 points. [06]

**Action:** Warn earlier for evening flights.

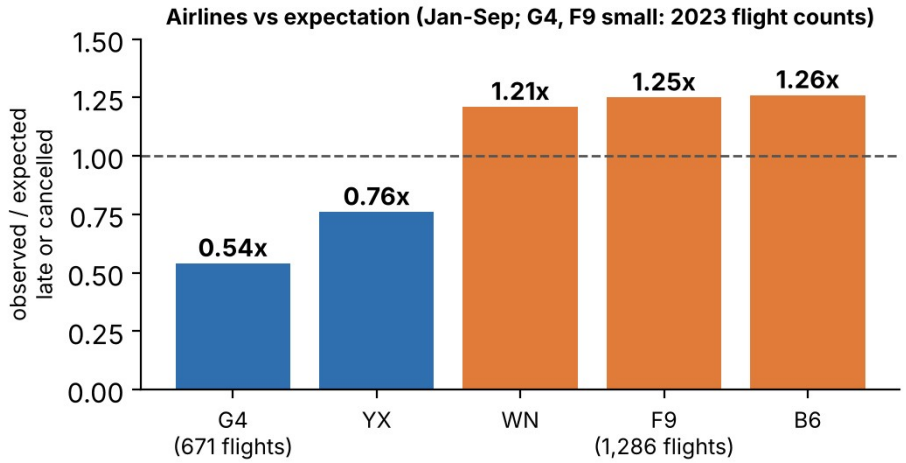
**Action:** Add a bad-day rule.

**Action:** Watch the airport live.

Sources: notebook 02 (Jan-Sep, day-bootstrap intervals), notebook 06 (importance)

# Airlines differ on bad days, and the weekday pattern was an accident of the calendar

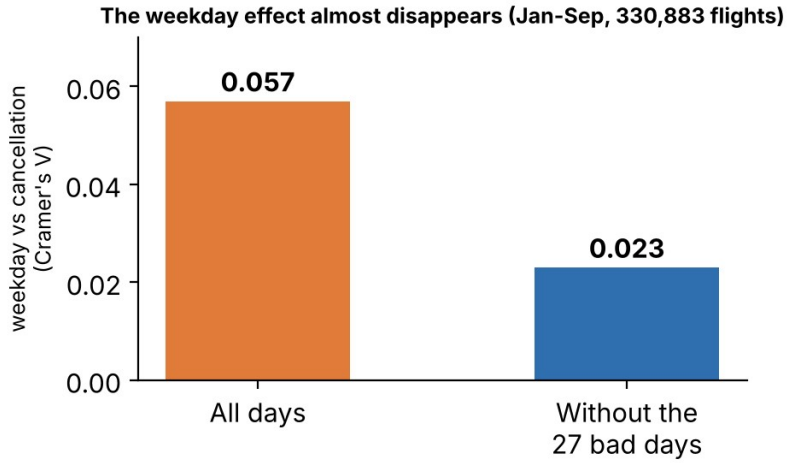
## 4. Airlines




On bad days **YX cancels 29%** and 9E 27%, **Delta 9%**. On ordinary days every big airline cancels 1.0% to 1.7%. [02]

**Action:** rebooking ready first for YX and 9E.

## 5. A strange pattern, explained




Thursday 1.5% vs Sunday 4.7% cancelled looks like a weekday effect. It nearly vanishes without the 27 bad days (**0.057 → 0.023**). [02]

**Action:** do not plan by weekday.

Sources: notebook 02 (Jan-Sep)

# Destinations fall into three groups, each with its own first action

## Short-haul

57 airports · 101,287 flights (Jan-Sep) [03]

**21.2% late**

Most cancellations (4.3%)

**Action:** Have rebooking ready first.

## Main long-haul

45 airports · 226,896 flights (Jan-Sep) [03]

**27.8% late**

Late 15% in the morning, 42% in the evening

**Action:** Warn earlier on evening flights.

## Chronic delays

7 airports · 2,374 flights (Jan-Sep) [03]

**53.1% late**

Mean delay 41.5 minutes

**Action:** Always warn and check connections.

The groups are a guide, not a certainty (silhouette 0.322). [03]

Sources: notebook 03 (k-means, Jan-Sep)

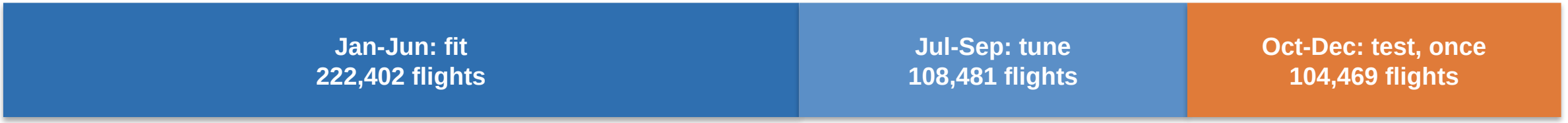
# Each score uses only what is known two hours before, and is tested once on months it never saw

**Known two hours before (used)**
schedule · route · flights that already left · previous flight of the plane · weather · plane details [05]

**T = 2 h before departure**

**After T (never used)**
actual times · delays · cancellation flags [05]

## 4 automated leak tests, all pass [05]

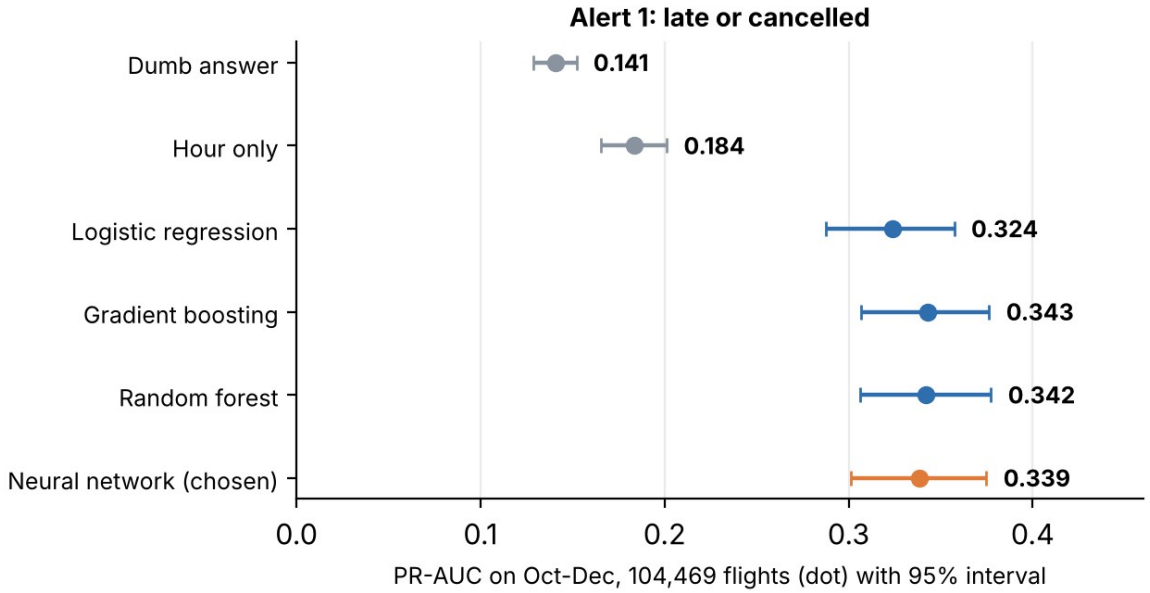


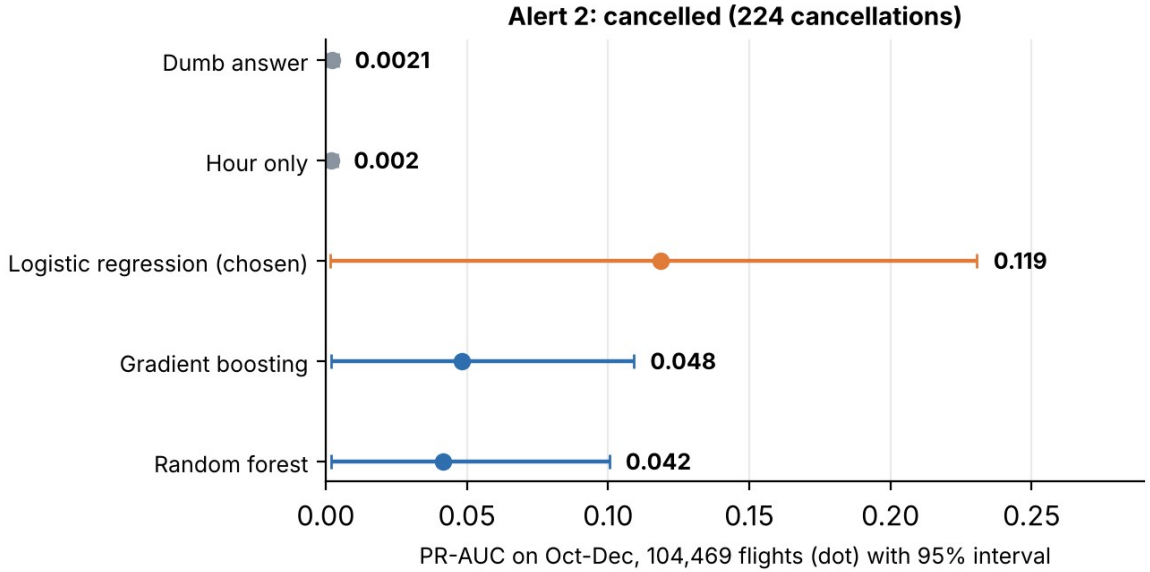

Final model refit on Jan-Sep (330,883 flights). [06]

**Why not accuracy?** "Nobody is late" is **85.9% accurate** and finds nothing. We compare every model with that dumb answer using PR-AUC: **0.1406** (Alert 1), **0.0021** (Alert 2). [06]

Sources: notebook 05 (leak tests), notebook 06 (protocol)

# Both alerts beat the dumb answer; the models are close, and cancellation is uncertain




### Alert 1

Neural network, best on validation. On test all four models are close. Without airport state: −0.055. [06]

### Alert 2

Logistic regression. Very wide interval: 224 cancellations, 115 on one day. [06]

### Weather choices

Strict weather +0.0019 and literal join +0.0005: both inside the interval. [06]

Sources: notebook 06 (final tables, ablation rows)

# Alert 1: about 120 flights flagged a day, and more than one in three really goes wrong

**123 a day**

flights flagged (11,275 in 3 months; cut-off 0.603) [07]

**38.0%**

of flags really late or cancelled (interval 34.7 to 41.4); random: 14.1% [07]

**29.2%**

of problem flights caught (interval 24.1 to 33.8); 2.7x random [07]

**0.339**

PR-AUC (interval 0.301 to 0.375) vs 0.141 for the dumb answer [06]

**What to do**

Message the client two hours before and check connections. A person confirms the flight status first.

**Watch out**

Weak in the morning (5.3% of problem flights caught) and at Newark (19.1%). [06]

**Re-check when**

Weekly recall is below 13% two weeks running (weeks with 100+ cases: 13.1% to 49.6%). [07]

**Question:** more than 15 minutes late or cancelled, asked two hours before departure. **Model:** neural network trained on Jan-Sep 2023 flights. **Why recall counts:** a false alarm costs one message; a missed delay leaves a client unwarned.

**Finding:** airport state two hours earlier is worth more than all weather columns (without it: −0.055 PR-AUC). [06]

Sources: notebook 07 (card 1, Oct-Dec test, 104,469 flights), notebook 06

# Alert 2: about 5 flights a day to prepare rebooking for, catching 41.5% of cancellations

**5.3 a day**

flights flagged (484 in 3 months; cut-off 0.136) [07]

**19.2%**

of flags really cancelled (interval 0.6 to 25.7); random: 1 in 500 [07]

**41.5%**

of cancellations caught: 93 of 224 (interval 1.0 to 59.9); 90x random [07]

**0.119**

PR-AUC (interval 0.002 to 0.231) vs 0.0021 for the dumb answer [06]

**What to do**

Offer a rebooking before the airline cancels. Never auto-rebook.

**Watch out**

Missed all 39 Newark cancellations. 115 of 224 fell on one day (18 Dec). [07]

**Re-check when**

Recall is below 23% in a bad-day block (half of the 46.8% seen in validation). [07]

**Question:** will the flight be cancelled, asked two hours before departure. **Model:** logistic regression trained on Jan-Sep 2023 flights. **Why recall counts:** a false alarm costs one message; a missed cancellation leaves a client stranded.

**Use it as a ranking:** the top 1% of scores catches 47.3% of cancellations (47x random). [07]

Sources: notebook 07 (card 2, Oct-Dec test, 104,469 flights), notebook 06

# The limits are named, and a weekly check tells us if the alerts stop working

## What it cannot do

- **Mornings:** only 5.3% of problem flights caught. [06]
- **Newark:** 19.1% for Alert 1; none of the 39 cancellations for Alert 2. [06]
- **Fairness:** flags JetBlue most; Alaska, American and Endeavor less than their real risk. [06]
- **Drift:** late or cancelled fell from 25.7% to 14.1%. [01]
- **Weather** is observed, not forecast (strict version: +0.0019). [06]

## Weekly check and stop rules

| | Watch weekly | Stop when |
|---|---|---|
| **Alert 1** | share of problem flights caught | below **13%** two weeks running |
| **Alert 2** | cancellations caught (4-week blocks) | below **23%** in a bad-day block |

**Always:** a named person confirms the flight status before any client contact. [07]

These rules rest on 14 weeks of data. They are **tentative**: review after the first quarter. [07]

Sources: notebooks 01, 06, 07

# Built with AI and checked by hand: every number traces back to the code

## How AI (Claude and Gemini) helped

- Wrote and debugged code (memory crash, tie bug, divide-by-zero)
- Helped plan the decisions
- Helped word the explanations and slides

## What we checked ourselves

- The 15 metric check values, asserted in code [04]
- 4 leak tests on every feature [05]
- Test months scored once
- All 7 notebooks re-run top to bottom: same numbers

Every number has a tag [01]-[07] and a line in the README ledger.

Data: nycflights23 (C. Ismay, S. P. Couch, H. Wickham, 2025, CC0).

Sources: README.md, notebooks 01-07

# Three yeses and we start a careful pilot

1. **Approve Alert 1:** about 120 messages a day [07], a person confirms each.
2. **Approve Alert 2:** about 5 flights a day [07] for rebooking readiness, never automatic.
3. **Approve the weekly check** with one named owner.

**To make it stronger**

a real weather forecast

the time each cancellation became known

a bad-day trigger [02]

**What we need from you**

how many messages a day the team can send

who confirms the flight status

how long the pilot runs

Sources: notebooks 02, 07